Validation and AI Use - TSLA Project 1

Decision, version, and as-of boundary

Decision/user: watch/defer TSLA for a buy-side equity-research committee with no current position. **Run command:** `py -3 app.py --write-output`. **Financial-statement and bridge as-of date:** 2025-12-31. **Screen/market-reference date:** 2026-09-10, using the month-end August 2026 checkpoint retained in early local notes. **Edition A disclosure:** the original baseline and receipt are unavailable; the companion research-evolution document contains a clearly labeled retrospective recollection, not a recreated original. **Version/commit:** no Git executable or commit history was available in this workspace; create a commit before submitting and enter it here: **STUDENT MUST ENTER COMMIT ID**.

Data and convention ledger

Material input	Source	As-of	Definition/units	Treatment	Risk
EBIT	Tesla 2025 10-K p. 50	FY2025	$4,355M	Base-year operating anchor	Includes potential normalizations
Cash/investments	Tesla 2025 10-K p. 49	2025-12-31	$16,513M / $27,546M	Gross addition in initial bridge	Operating-liquidity /tax policy unresolved
Debt/finance leases	Tesla 2025 10-K Notes 9-10	2025-12-31	$8,400M	Financing claim deducted	Consolidation policy must match EV
Diluted shares	Tesla 2025 10-K p. 50	FY2025	3,528M	DCF per-share denominator	Weighted average, not current count
Revenue/margin paths	Analyst assumptions	FY2026-2030	5%-7% growth; 6%-10% margin	Bounded exposed controls	Not management guidance
WACC/terminal growth	Analyst assumptions	Model run	10.5% / 2.5%	DCF discount/terminal convention	Terminal value is 80.6% of EV

Validation register

Claim/output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
DCF arithmetic	Terminal value discounted at wrong period	Lab 05 training fixture should equal $27.4974/share within $0.01	$27.49737; pass	Accepted	`src/model.py: known_answer_test`, visible output JSON
Statements articulate	Balance sheet or cash flow does not tie	Annual BS error = 0; ending cash equals prior cash plus CFO less capex	All five years pass; max error 0.00	Accepted	visible output JSON `checks`
Bridge	Incorrect claim treatment	Recompute EV + cash + investments - debt/finance leases - NCI/redeemable NCI	Pass	Accepted subject to cash policy	visible output JSON `checks`

Claim/output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
DCF direction	WACC/growth sign reversal	Higher WACC lowers value; higher terminal growth raises value	Both pass	Accepted	visible output JSON `validation`
Highest-risk input	Incoherent terminal relation silently accepted	Set terminal growth to 11% with WACC 10.5%; expect failure	Fails loudly; no valuation	Accepted	`app.py --interactive` and JSON validation
Source definition	Mix current shares with weighted-average shares	Model must use one stated convention throughout	3,528M used throughout	Qualified	evidence ledger
Comps/transactions	Unsupported peer multiple affects action	Require point-in-time market/financial source packet before use	Not complete; omitted from recommendation	Qualified limitation	decision memo

Locked Changed-Input Record - human prediction first; AI closed during execution

This record must be authored by the student before the run. It cannot be truthfully completed by an AI after the fact. The engine has a ready-to-run margin test, but you must make the prediction in your own words, timestamp it, and commit/snapshot it before executing.

Precommit timestamp	Repository commit	Material input	Old -> new value/units	Expected output direction	Expected decision effect
STUDENT MUST COMPLETE BEFORE RUN	**STUDENT MUST COMPLETE**	EBIT margin shift	0.0pp -> +1.0pp	**STUDENT'S AI-OFF PREDICTION**	**STUDENT'S AI-OFF PREDICTION**

Before/after output	Actual decision effect	Prediction reconciliation	Failure diagnosis or no-change reason	Evidence path
$22.09 -> $25.06 per share	Watch/defer remains unchanged because result remains far below retained market reference	**STUDENT MUST EXPLAIN AFTER RUN**	No action change because the tested improvement is insufficient	`output/visible_output.json` validation section

Named AI-use register

Tool/model surface and exposed version	Interaction date	Material task	Output/claim used or considered	Independent check	Disposition	Effect on decision
OpenAI Codex, GPT-5 surface (UI version not exposed)	2026-10-19	Drafted a standard-library model and documentation structure	FCFF/three-statement implementation and document organization	Lab 05 known-answer fixture; source-linked input ledger; articulation, bridge, and monotonicity tests	Modified and accepted with limitations	Enabled reproducible tool; did not establish finance facts
OpenAI Codex, GPT-5 surface (UI version not exposed)	2026-10-19	Suggested a margin changed-input test	+100bp margin should increase FCFF/value	Independent execution: $22.09 -> $25.06/share	Accepted as a mechanical test, not as evidence for forecast margin	No action change

Limitations, monitoring, and kill/escalation rules

Do not rely on the DCF as a point estimate. Refresh peer and transaction evidence, reconcile the latest filing, define excess operating cash, reconcile a current diluted share count, and revisit regulatory-credit, legal, digital-asset, and SpaceX-investment treatment. Reopen initiation only after two consecutive filings support a credible operating-margin/cash-flow path and the bridge is refreshed. Withdraw the valuation conclusion if the bridge fails or input classifications change materially.